G. Can You Split the Array?

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given an array containing N positive integers. Your task is to partition the array into at most K subarrays so that the maximum sum among all subarrays is as small as possible.

Input

The first line contains two integers N $(1 \leq N \leq 2 \times 10^5)$ and K $(1 \leq K \leq N)$, the size of the array and the number of subarrays in the partition.

The next line contains N integers $a_1, a_2, \ldots, a_N$ $(1 \leq a_i \leq 10^6)$, the elements of the array.

Output

Print one integer: the minimum possible largest subarray sum.

Examples

input

```
5 3
1 2 3 4 5
```

output

```
6
```

input

```
5 3
1 5 1 1 5
```

output

```
6
```

Note

Brief Explanation for First Sample: There are many ways to partition the array

- The optimal/best partition is $[1, 2, 3], [4], [5]$. The maximum sum for this partition is 6. This is because
  - the sum of the first subarray $[1, 2, 3]$ is $1 + 2 + 3 = 6$;
  - the sum of the second subarray $[4]$ is 4; and
  - the sum of the third subarray $[5]$ is 5.
- Another partition is $[1, 2], [3, 4], [5]$. The maximum sum for this partition is 7. This is because
  - the sum of the first subarray $[1, 2]$ is $1 + 2 = 3$;
  - the sum of the second subarray $[3, 4]$ is $3 + 4 = 7$;
  - the sum of the third subarray $[5]$ is 5.

Since we want to find the smallest possible value for the maximum sum, the first partitioning ($[1, 2, 3], [4], [5]$) is the best. The answer is 6.

Problem Extension: What if we allow negative numbers? How would the solution change?
